

Eranga Madapathage

Software Engineer (Python & Node.js) | AI & LLM Systems

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GitHub: <https://github.com/erangamdw> | Portfolio: erangamdw.com | Right to Work in the UK

Software Engineer with 6+ years of experience building scalable backend systems and AI-powered applications. Specialised in Python, FastAPI, Node.js, and modern React-based architectures, with hands-on experience integrating LLMs such as OpenAI and LangChain into production systems. Strong background in designing RESTful APIs, real-time systems, and intelligent automation solutions. Holds an MSc in Data Science and Artificial Intelligence, with a focus on applied AI systems.

Work experience

Software Engineer – Efito Solutions (Pvt) Ltd, Sri Lanka

Aug 2023 – Sep 2024

- Designed, developed, and maintained backend services using Python (FastAPI) and Node.js, supporting 10K+ active users across multiple production applications.
- Built and optimised scalable RESTful APIs, implementing authentication and authorization using JWT, session management, and role-based access control (RBAC).
- Integrated LLM-powered features for the Compatible1 platform using OpenAI APIs, enabling AI-assisted workflows and intelligent automation in real-world user flows.
- Developed backend logic for AI-enhanced call management, integrating Python/Node services with Socket.io and ZegoCloud for real-time communication.
- Improved backend performance to achieve <200ms average response times through query optimisation, async processing, and efficient data modelling.
- Worked on full-stack features by developing React.js components and integrating them with secure backend APIs.
- Applied AI-assisted development practices (GitHub Copilot, prompt engineering) to accelerate delivery while maintaining code quality.
- Mentored 5+ junior developers, conducted code reviews, and promoted backend best practices, improving delivery speed and overall code quality.
- Contributed to a food bank management platform, optimising backend APIs and improving system performance by ~25%.

Software Engineer & Technical Support Engineer – LAYOUTindex UK Ltd, Stevenage (SL Development Centre)

May 2021 – Jun 2023

- Developed and maintained backend and full-stack applications using Python, Node.js, React.js, and React Native.
- Designed and secured REST APIs, implementing authentication, access control, and backend validations for enterprise and consumer-facing systems.
- Worked on systems delivered to UK-based clients, ensuring solutions met production, security, and reliability standards.
- Diagnosed and resolved backend, database, and infrastructure issues, maintaining ~99% uptime across critical platforms.
- Collaborated with cross-functional teams to design scalable, secure, and maintainable backend architectures.

Associate Software Engineer – EMP (Pvt) Ltd, Sri Lanka

Feb 2018 – Mar 2021

- Supported ERP (SAP + MS SQL Server) enhancements, improving workflow efficiency by 15%.
- Developed and maintained internal web applications and backend services using FastAPI and NodeJS, supporting business workflows, automation, and internal data management needs.

Education

- **MSc. Data Science & Artificial Intelligence** - Edge Hill University, Lancashire (2024 -2025)
- **BSc (Hons) Computer Science (Software Engineering)** – University of Wolverhampton – First Class (2022 -2023)
- **Higher National Diploma in Computing (Software Engineering)** – Pearson BTEC (2018 -2020)

Core Technical Skills

- **Programming:** Python, JavaScript, TypeScript
- **Python Frameworks:** FastAPI, Flask
- **AI & LLMs:** OpenAI GPT models, Claude (Sonnet), LangChain, Hugging Face Transformers, Ollama (LLaMA, Mistral, Phi)
- **Machine Learning:** TensorFlow, PyTorch, Scikit-learn
- **AI Techniques:** Prompt engineering, RAG pipelines, AI workflow automation, model evaluation & inference optimisation
- **Backend & APIs:** RESTful API design, microservices architecture, async services
- **Databases & Data Stores:** PostgreSQL, MongoDB, MySQL, SQLite, Redis, Vector Databases (Pinecone, Chroma)
- **Cloud & DevOps:** Docker, AWS (EC2, Lambda, S3, API Gateway, CloudWatch), CI/CD pipelines
- **Testing:** pytest, unit & integration testing
- **Version Control & Tools:** Git, GitHub, GitLab, Hugging Face Hub, OpenAI APIs
- **Supporting Technologies:** Node.js, Express.js, React.js, React Native, Flutter

Projects

InstaLeaf – AI-Powered Plant Disease Detection

- Designed and implemented a Python-based backend for plant disease classification using TensorFlow / PyTorch.
- Optimised deep learning models for CPU-based inference, focusing on performance and deployment efficiency.

Compatible1 – AI-Enhanced Communication Platform

- Integrated LLM-powered features using OpenAI APIs to enable AI-assisted workflows and intelligent automation in real user scenarios.
- Worked with session handling, request validation, and usage control to ensure safe and scalable AI feature usage.

Smart Brain Health App – AI-Driven Brain tumour detection App

- Designed and developed a mobile application that analyses brain MRI images using AI-based models to support early detection of brain tumours.

HireLens AI – AI-Powered Interview and Hiring Intelligence Platform

- Built a full-stack AI platform for candidate evaluation, job matching, and interview intelligence using FastAPI, Next.js, and LangChain
- Developed a RAG-based system enabling context-aware insights over CVs and job descriptions
- Implemented LLM-powered features including interview question generation, skill gap analysis, and personalised candidate feedback
- Developed AI-driven CV-to-job matching and ranking, supporting data-driven hiring decisions

DocMind AI – AI Knowledge System for Multi-Source Document Intelligence using RAG

- Built a full-stack AI system for natural language querying over documents using RAG
- Developed backend services with FastAPI, LangChain, and Qdrant for semantic search and LLM-powered responses
- Designed secure APIs and a React dashboard for managing multiple knowledge bases

Achievements

- Research paper publication in IEEE Xplore "Smart Health App for Identifying Brain Tumors" | 2023 IEEE 8th International Conference for Convergence in Technology (I2CT) Available at: [Link to the paper](#)